20 Show that $\frac{\sqrt{180} - 2\sqrt{5}}{5\sqrt{5} - 5}$ can be written in the form $a + \frac{\sqrt{5}}{b}$ where a and b are integers.

√180 = √(36 × 5) = 6√5, so numerator = 4√5.

4√5/[5(√5 − 1)] × (√5 + 1)/(√5 + 1)

= (20 + 4√5)/[5(5 − 1)]

= (20 + 4√5)/20 = 1 + √5/5.

4 marks total
Simplify √180
Use conjugate
Expand numerator/denominator
1 + √5/5

(Total for Question 20 is 4 marks)

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